

Can We Build a VNN-COMP Witness Checker? A SONNX-Related Proposal

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DISCLAIMER: what follows is a draft, please excuse the occasional typo.

1 The Problem

The international competition of neural network verification¹ (VNN-COMP) has a long standing issue. To understand it, let us first look at a classic neural network verification query, i.e. adversarial robustness:

$$\exists x', \|x - x'\| \leq \epsilon \implies \arg \max\{f(x)\} \neq \arg \max\{f(x')\} \quad (1)$$

Given a specific reference input x and a neural network f , the query asks whether there exists a small adversarial perturbation x' that changes the output class.

The tools that participate in VNN-COMP can solve queries like the above for networks with up to 100M parameters. Their general workflow is depicted in Figure 1. From a very high level perspective, the tool receives as input a neural network in ONNX format, and a property specification in VNN-LIB format. It then runs some custom decision procedure, after which it declares the query UNSAT (e.g. there is no adversarial perturbation) or SAT. In the latter case, a counterexample x' is returned, i.e. just the network input, not a full execution trace.

Most of the benchmarks used during the competition (and most of the papers in the literature) assume that the neural network has real-valued semantics, i.e. $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$. However, the large majority of tools ignore this fact and use instead a hacky combination of unsound decision procedures and floating-point arithmetic.²

As a result, *some of the counterexamples are not reproducible*. In fact, tools occasionally disagree on the UNSAT vs SAT verification results.³ As you can

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¹<https://vnn-comp.github.io/>

²E.g. the tool might run a incomplete falsifier on an arbitrary floating-point implementation. If that does not find a counterexample, the tool might switch to verification procedure that is theoretically correct for real-valued computation but has been unsoundly implemented in floating-point.

³Currently, the competition decides the ground truth by majority voting.

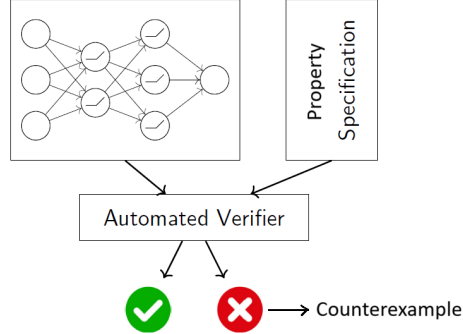


Figure 1: High-level workflow of a neural network verifier.

imagine, this situation leads to very heated discussions about the scoring system and how such “ambiguous” instances should be adjudicated.⁴

2 The Solution

What the competition needs is a *witness checker*. I.e. a tool that can take the counterexample input, execute it on a reference implementation, and check whether its output actually violates the property. That way, the witness checker can become the “final referee” in the competition, and score all SAT verdicts against the actual ground truth.

The question for us is whether we can generate such reference implementation automatically from the SONNX specification. Importantly, the reference implementation must support *real-valued computation*, since VNN-COMP benchmarks contain real-valued queries in > 99% of the cases.

Libraries that support arbitrary precision with rational numbers, or propagate floating-point intervals around the real-valued semantics exist. Can we easily integrate them into our current Why3 workflow?

3 Strategic Considerations

The above proposal will require some amount of work. Why should we bother? I see three reasons to do so:

- The witness checker will be a concrete example of the impact of SONNX on the larger research community, even if it is only a small step.
- The VNN-COMP benchmarks cover a relatively large subset of ONNX operators, as shown in Figure 2. Having a user base that cares about our witness checker will provide an incentive to slowly support all of them.

⁴<https://github.com/VNN-COMP/vnncomp2026/issues/2>

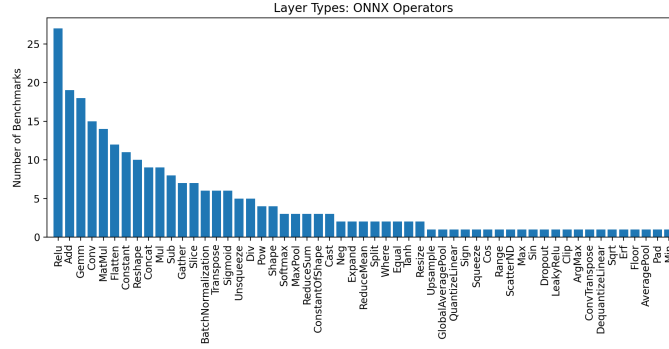


Figure 2: How many VNN-COMP benchmarks does a particular ONNX operator appear in?

- The witness checker will bring SONNX even more to the attention of the VNN-COMP community, which might help with recruitment.

4 Timeline

The next edition of VNN-COMP will begin around April-May 2027, with the final competition run happening around June-July 2027. Having a prototype before hand would allow us to "live test it" during the actual competition.